INCH-POUND
MIL-PRF-32383/4A(CR)
01 January 2022

SUPERSEDING
MIL-PRF-32383/4(CR)
6 March 2013

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED, CONFORMAL WEARABLE BATTERY (CWB), BB-2525

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-32383.

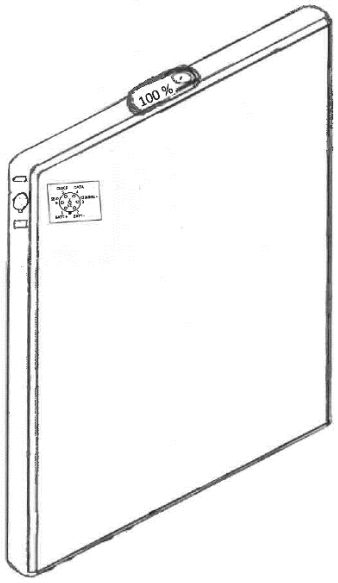


FIGURE 1. M32383/4A-III (BB-2525) CWB (see FIGURE 11 for dimensions)

NOTES: Depicted battery is representative of a potential solution but is not a requirement within this document.

Unless otherwise specified, paragraph numbers are in reference to MIL-PRF-32383. Note that the requirement and verification paragraphs included in this document that are in addition to or modified from MIL-PRF-32383 are highlighted in **BOLD**.

For clarity, MIL-PRF-32383 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32383/4A tables are labeled with lower case Roman numerals (e.g. ix).

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APPLICABLE DOCUMENTS.

1. Government documents (2.2).

Add:

MIL-STD-461 Department of Defense Interface Standard: Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

2. Other Government documents, drawings, and publications (2.2.2).

Add:

ADS-37A-PRF Electromagnetic Environmental Effects (E3) Performance and Verification Requirements

(Copies of this document may be obtained online at https://www.avmc.army.mil/Portals/51/Documents/TechData%20PDF/ADS37APRF.pdf)

NWPAN-WP-01112013 Nett Warrior Interconnect Architecture White Paper

(Available from: Product Manager Ground Soldier Systems (PdM GSS) Attn: SFAE-SDR-CCS, 5966 12th Street, Building 1024, Ft. Belvoir, VA. 22060) – Approved Nett Warrior compatible connectors.

REQUIREMENTS.

Add: Specification requirements and test methods: The following requirements and test methods of MIL-PRF-32383, identified therein as “when specified”, are applicable or not applicable as indicated below:

TABLE i. *“When specified” requirements*

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Additional material requirements	*3.3.7*	*4.7.1.9*	*No*
Connector pull force	3.4.4.1.1.1	4.7.1.3.1	Yes
Connector insertions	3.4.4.1.1.2	4.7.1.3	Yes
Multilayer finish	3.4.4.1.3	Certification	Yes
Gold plating	*3.4.4.1.4*	*Certification*	*No*
USB power port connector(s)	3.4.4.1.5	4.7.2.17	Yes
USB type A connectors(s)	*3.4.4.1.5.1*	*4.7.2.17*	*No*
USB type C connectors(s)	3.4.4.1.5.2	4.7.2.17	Yes
Surface friction	3.4.5.6	4.7.1.8	Yes
Item unique identification (IUID)	3.4.6.1.1	4.7.1.1	Yes

NOTES

[1] As modified herein, both mated and unmated configurations.

[2] As specified herein.

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TABLE i. *"When specified" requirements* - Continued

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Additional physical characteristics requirements	*3.4.8*	*4.7.1.10*	*No*
Inrush current	3.5.5	4.7.2.12	Yes
Charge enable	*3.5.6.2*	*4.7.2.6*	*No*
Display, non-smart, always on	*3.5.8.1.1*	*4.7.2.14.1*	*No*
Display, non-smart, user activated	*3.5.8.1.2*	*4.7.2.14.1*	*No*
Display, smart	3.5.8.2	4.7.2.14.2	Yes
SMBus[1]	3.5.9.1	4.7.2.15.1	Yes
USB power delivery	*3.5.9.2*	*4.7.2.15.2*	*No*
Dual-role power device	3.5.9.2.1	4.7.2.14.2	Yes
Concurrent charge/discharge	3.5.11	4.7.2.16	Yes
Additional electrical requirements	*3.5.12*	*4.7.2.18*	*No*
Temperature shock high altitude high opening (HAHO) simulation	3.6.3	4.7.3.5.1	Yes
Temperature shock high altitude low opening (HALO) simulation	3.6.3	4.7.3.5.2	Yes
Vibration, loose cargo transportation	3.6.5	4.7.3.7.1	Yes
Immersion, shallow[1]	3.6.6	4.7.3.8.1	Yes
Immersion, deep	*3.6.6*	*4.7.3.8.2*	*No*
Transit drop, normal	*3.6.7*	*4.7.3.9.1*	*No*
Transit drop, severe	3.6.7	4.7.3.9.2	Yes
Drop while mated[1]	3.6.7	4.7.3.9.2	Yes
Impact	3.6.9	4.7.3.11	Yes
Solar radiation	3.6.10	4.7.3.12	Yes
Salt fog	3.6.11	4.7.3.13	Yes
Electrostatic discharge (ESD)	3.6.12	4.7.3.14	Yes
Chemical resistance	3.6.13	4.7.3.15	Yes
Rapid decompression	3.6.14	4.7.3.16	Yes
Explosive atmosphere	3.6.15	4.7.3.17	Yes
Combined thermal, altitude, and humidity	3.6.16	4.7.3.18	Yes
Additional environmental requirements[2]	3.6.17	4.7.3.19	Yes
Nail penetration, cell	3.7.1.2	4.7.4.4	Yes
Crush, cell	3.7.1.3	4.7.4.5	Yes
Inter-cell protective devices	3.7.2.6	Certification	Yes
Redundant safety protection	3.7.2.7	Certification	Yes

NOTES

[1] As modified herein, both mated and unmated configurations.

[2] As specified herein.

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TABLE i. *"When specified" requirements* - Continued

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Projectile	3.7.2.8	4.7.4.11	Yes
Nail penetration, battery	3.7.2.9	4.7.4.12	Yes
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5	Yes
Weapons characterization	3.7.2.11	4.7.4.13	Yes
Additional safety requirements	*3.7.3*	*4.7.4.14*	*No*
Electromagnetic interference	3.8	4.7.7	Yes
Additional transportation requirements	*3.9.1*	*4.7.8*	*No*
Reliability	3.11	4.7.2.4.1	Yes

NOTES

[1] As modified herein, both mated and unmated configurations.

[2] As specified herein.

3. Certifications (3.2.1).

Delete: The following line in TABLE II of MIL-PRF-32383:

REQUIREMENT	PARAGRAPH
Connector pull force	3.4.4.1.1.1

Add: The following line in TABLE II of MIL-PRF-32383:

REQUIREMENT	PARAGRAPH
Battery sleep mode	3.6.8.1

4. Dimensions and weight (3.4.3).

Width shall be 7.60 +/- 0.06 inches.
Height shall be 8.65 +/- 0.05 inches.
Depth shall be no less than 0.50 inches and not greater than 0.70 inches.
Weight shall be no greater than 2.6 pounds.

All other dimensions shall be accordance with FIGURE 11.

5. Connector (3.4.4.1.1).

Certification of the following is required:

Nett Warrior Connector: Glenair connector Part Number: 8070-1299-ZNU6-7DY (printed circuit board tails or solder cup), or TE Connectivity Part Number: 2828420-2 (7 position). The primary (master) keyway shall be oriented downward, facing the flat terminal contacts.

The connector shall provide a shell to shell shield impedance of not greater than 4 milliohms when mated with any of the mating connectors listed in NWPAN-WP-01112013.

The ground (Socket 2 V- Battery Ground) shall be terminated just within the battery pack to the

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shield shell and shall have no more than 4 milliohms impedance between ground socket 2 and the shield shell. When connected with accordance with FIGURE 13, and Cable G, the impedance of the ground socket 2 shall be not greater than 15 milliohms.

Sockets 1, 2, and 3 shall be capable of carrying not less than 5 amps continuous each.

Connector shall be capable of mating with plugs having the 6-6 or 6-7 pin configuration.

The connector shall be located on the left side of the battery at 7.89 inches from the bottom of the battery +/- 0.5 inches to allow for USB connector spacing. Refer to FIGURE 11.

The pinout for the connector shall be as follows in TABLE ii with socket numbering depicted in FIGURE 2.

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TABLE ii. *Nett Warrior connector pinout*

Socket	Description	Terminal markings
1	V+ Battery Voltage	Batt +
2	V- Battery Ground[1]	Batt -
3	Charge +[2]	Charge +
4	SMBus Data	Data
5	SMBus Clock	Clock
6	SMBus ID ('T-pin')[3]	SB ID
7	No Connection (N/C)[4]	N/C

NOTES

[1] Requires internal termination of Ground to Shield Ground

[2] Charge + shall be input only

[3] Safety Signal in accordance with Smart Battery Data Specification Revision 1.1 Safety Signal Hardware Requirements

[4] Shall not be used for any electrical connectivity or for grounding

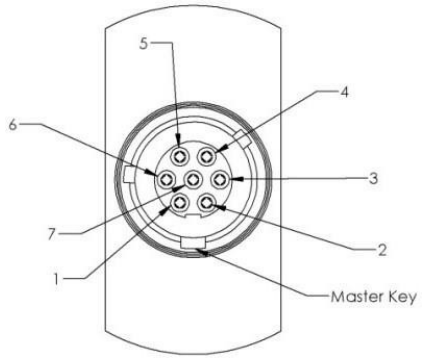


FIGURE 2. Nett Warrior connector pinout

6. Connector pull force (3.4.4.1.1.1).

The Nett Warrior connector pull force shall be 12.0 +/- 4.0 pounds.

7. Flat terminal (3.4.4.1.2).

The charge terminals shall be as shown in FIGURE 11, Charge Terminal Pinout Assignment. All four contacts shall be recessed 0.020 +/- 0.005 inches. The contacts shall support charging at a minimum of 7.5 Amps.

Delete: Battery terminals, including data and charge terminals, shall be of sufficient strength to withstand 250 connections without damage that would interfere with connections and shall maintain a fully functional interface.

Add: Charge terminals shall sustain not less than a 20 pound force each for 10 minutes without degradation to the battery contacts from a charge contact pin.

The CWB shall contain at least one USB 3.0 Type-A connector. The USB Connector shall be

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located within the Connector region of FIGURE 11.

8. Battery case (3.4.5).
Delete: Acceptable ratings include: HB; V-0; V-1; V-2; 5VA; 5VB; VTM-0; VTM-1; and VTM-2.

Add: Acceptable ratings include V-0, 5VA, or 5VB.

9. Battery case vent (3.4.5.3).
Pressure release device(s) shall be located on the CWB front face in accordance with FIGURE 11.

10. Surface friction (3.4.5.6).
Batteries shall slide easily in and out of their pouch. When tested as specified in 4.7.1.8 batteries shall have a coefficient of static friction of not greater than 0.40 against a Cordura® fabric containing meta-aramid, nylon and cotton in a blend ratio of 43/30/27 percent by weight.

11. Marking (3.4.6).
Add:
Warning markings shall be not less than a 12 point font; other markings shall not be less than a 10 point font.

12. Warning marking (3.4.6.2).
Delete:
DO NOT STORE ABOVE 122°F (50°C),

Add:
DO NOT STORE ABOVE 160°F (71°C),

13. Instructions / notes (3.4.6.5).
Add:
The following charging instructions shall be marked on each battery:

CHARGE CONSTANT VOLTAGE AT 16.8V AT 5 AMPS MAX THROUGH NETT WARRIOR CONNECTOR.
BATTERY TEMPERATURE SHOULD BE BETWEEN -4°F (-20°C) AND 122°F (50°C) DURING CHARGE.
LOWER TEMPERATURE CHARGING [BELOW 32°F (0°C)] IS PERMITTED, BUT RUNTIMES WILL BE LOWER.

(Note: if charge voltage is different than 16.8 VDC, voltage should be stated on battery charging instructions.)

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14. Simplified battery label (3.4.6.6).
The simplified battery label location shall be located on the same surface as the charging instructions and shall measure 2.5 x 1.5 inches ± .1 inches.

15. Additional markings (3.4.6.9).
The Nett Warrior Connector marking shall be as shown in FIGURE 3 and orientated as in FIGURE 11. The orientation shall have the master key way facing down towards the bottom of the battery pack.

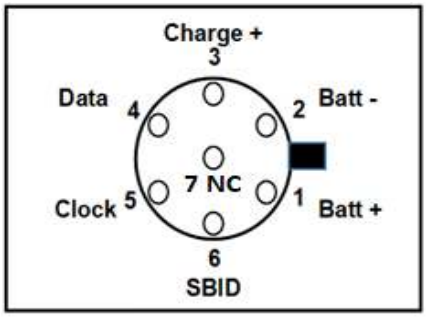


FIGURE 3. Nett Warrior connector marking

16. Battery condition for shipping (3.4.7).
Add:
The instruction card that shall contain instructions for operation, charging, and maintenance of the battery to include long term storage/shipping requirements per (Safety Data Sheet, class of shipping, and shipping markings).

17. Dust cap or cover (3.4.7.1).
Add:
The Nett Warrior Connector dust cap does not need to be permanently tethered to the CWB, however, if it is, it must not interfere with any other (i.e. USB ports) individual insertion or removal while wearing Army tactical gloves. If the Nett Warrior connector is supplied as tethered protective covers it shall remain within the battery tolerances specified in FIGURE 11.

18. Battery voltage (3.5.2).
Battery shall operate within the range of 10.0 volts to 20.0 volts.

19. Capacity (3.5.3).
Cutoff voltage: 10 volts.

When subjected to the specific test listed below and discharged at the specified rate to minimal final voltage, the battery shall have the following minimum capacity:

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TABLE iii. *Test capacity and energy requirements*

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Capacity /Energy
Capacity discharge	3.5.3	4.7.2.3	See 4.7.2.3	148 Wh
Cycle life (125th cycle)	3.5.4	4.7.2.4	See 4.7.2.4	125 Wh
Reliability	3.11	4.7.2.4.1	18 W	125 Wh
High rate discharge	3.5.3	4.7.2.5	5 A	9.0 Ah
Low temperature discharge	3.5.3	4.7.2.7	18 W	75 Wh
High temperature discharge	3.5.3	4.7.2.8	18 W	140 Wh
Inrush current	3.5.5	4.7.2.12	See 3.5.5	148 Wh
Charge acceptance	3.5.10.1	4.7.2.9	18 W	148 Wh
High temperature charge acceptance	3.5.10.1	4.7.2.9	18 W	130 Wh
Low temperature charge acceptance	3.5.10.1	4.7.2.9	18 W	75 Wh
Retention of charge	3.5.10.2	4.7.2.10	18 W	135 Wh
Pulse discharge	3.5.3	4.7.2.11	See 4.7.2.11	7.0 Ah
Extreme low temperature discharge	3.6.1	4.7.3.2	18 W	75 Wh
Extreme high temperature discharge	3.6.1	4.7.3.3	18 W	130 Wh
Temperature shock (post-test)	3.6.3	4.7.3.5	18 W	130 Wh
Mechanical shock (post-test)	3.6.4	4.7.3.6	18 W	135 Wh
Vibration (during test)	3.6.5	4.7.3.7	18 W	148 Wh
Battery storage life	3.6.8	4.7.3.10	18 W	100 Wh
Impact	3.6.9	4.7.3.11	18 W	148 Wh
Flexibility	3.6.17.1	4.7.3.19.1	18 W	148 Wh
Over flexibility protection	3.6.17.2	4.7.3.19.1.1	18 W	148 Wh
Salt Fog	3.6.11	4.7.3.13	18 W	148 Wh
Electrostatic discharge	3.6.12	4.7.3.14	18 W	148 Wh
Conformability	3.6.17.3	4.7.3.19.2	18 W	148 Wh
Overcharge/electric leakage (post-test)	3.7.2.2	4.7.4.7	18 W	140 Wh
Short circuit protection (post-test)	3.7.2.3	4.7.4.8	18 W	135 Wh
High temperature temporary cut off	3.7.2.4	4.7.4.9	18 W	105 Wh
Electromagnetic interference	3.8	4.7.7	18 W	148 Wh
Rapid decompression	3.6.14	4.7.3.16	18 W	148 Wh
Explosive atmosphere	3.6.15	4.7.3.17	18 W	148 Wh
Combined temperature, altitude and humidity	3.6.16	4.7.3.18	18 W	148 Wh
Basic functionality	3.4.1	4.7.8.1	18 W	148 Wh

20. Cycle life (3.5.4).

When tested as specified in 4.7.2.4 batteries shall be capable of retaining the specified capacity during and after 125 cycles.

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21. Inrush current (3.5.5).

The discharge profile shall be as follows. Verify values in accordance with 4.7.4.12.

TABLE iv. *Inrush current*

STEP	Start time	End time	Current (A)
	(second:millisecond:microsecond)		
1	00:000:000	00:000:**400**	65
2	00:000:**400**	00:000:**915**	33
3	00:000:**915**	00:**025:**000	25
4	00:**025**:000	**01**:000:000	6.5
5	**01**:000:000	**02**:000:000	5.5
6	**02**:000:000	Until Cutoff Voltage	5.15

22. Charging interface (3.5.6.1).

Charge terminals shall be electrically protected against short circuit and shall not source battery current. The battery shall be capable of being charged via both the Nett Warrior connector and the charge terminal contacts. The battery shall be capable of “wake up” charging via both the Nett Warrior connector and the charge terminal contacts. Wake up charging via the Nett Warrior connector shall be in accordance with the Smart Battery Specified behavior for the Safety Signal. To perform wake up charging, via the charge terminal contacts, shall be performed by the battery accepting +5V between terminals 1 and 2 and responding to SMBus commands before charging voltage is provided.

TABLE v. *Charge terminal descriptions*

POSITION (Left to Right)	**DESCRIPTION**	**TITLE**
1	Charge Terminal, Positive	CHG +
2	Charger Terminal, Negative (Ground)	CHG -
3	SMBus Data	DATA
4	SMBus Clock	CLK

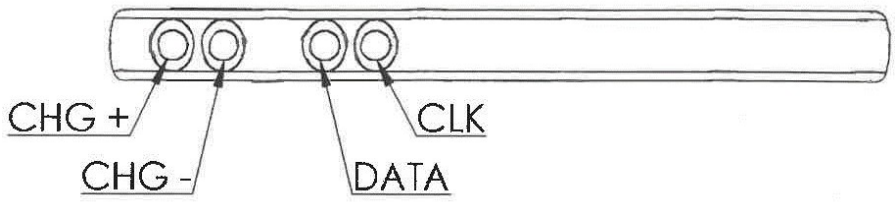


FIGURE 4. Charge terminal pinout assignment

23. Power consumption (3.5.7).

Add:

Power consumption shall be assessed post disconnect from active use or no load at a minute post any state of charge indicator (SOCI) button depression to illuminate / backlight or update display information.

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24. Display, smart (3.5.8.2).
Display shall be located as shown in FIGURE 11. The display shall have a font size of not less than 0.22 inches in height.

25. SMBus (3.5.9.1).
SMBus output from the connector should be sufficient to maintain signal integrity through the two (2) cables depicted as “C” and “D” in FIGURE 12.

26. Concurrent charge/discharge (3.5.11).
The battery shall be capable of being charged via pin 3 and simultaneously discharged via pin 1 of the Nett Warrior connector.

The battery shall be capable of being charged via the flat terminal contacts (see 3.4.4.1.2) and simultaneously discharged via pin 1 of the Nett Warrior connector.

27. Environmental requirements (3.6).
Delete: Each battery when subjected to specified environmental tests below shall show no:
Add: Each battery when subjected to specified environmental tests below shall meet be tested in accordance with 4.7.8.1 and shall show no:
Add: The following lines to TABLE VII of MIL-PRF-32383:

Tests	Requirement Paragraph	Test Paragraph
Flexibility	3.6.17.1	4.7.3.19.1
Over flexibility protection	3.6.17.2	4.7.3.19.1.1
Conformability	3.6.17.3	4.7.3.19.2

28. Battery sleep mode (3.6.8.1).
Manufacturer specific sleep mode shall not be utilized. If sleep mode is utilized, the battery shall not enter sleep mode within 48 hours of the last activity (i.e. button press, charge current, or battery load). Certification of the 48 hour delay is required.

29. Battery wake up (3.6.8.2).
The battery shall wake up from sleep mode after not greater than two seconds following battery activity. Battery activity shall also include when the SMBus Safety Signal is queried.

Add:
30. **Flexibility (3.6.17.1).**
Batteries shall be capable of flexing without sustaining physical or electrical damage. When tested as specified in 4.7.3.19.1 batteries shall flex with a 1 inch displacement in both directions on each axis and meet the requirements of 3.5.3 and 3.6.

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Add:

31. **Over flexibility protection (3.6.17.2).**

Batteries shall be capable of withstanding over-flex conditions without sustaining physical or electrical damage. When tested as specified in 4.7.3.19.2 batteries shall allow support of not less than 45 Lbs. bending force for intervals of a minimum of 5 seconds each without failure or damage to the battery. It shall meet the requirements of 3.5.3 and 3.6.

Add:

32. **Conformability (3.6.17.3).**

Batteries shall be capable of withstanding bending without sustaining physical or electrical damage. When tested as specified in 4.7.3.19.2 batteries shall withstand bending under not greater than 35 pounds of force around curved surfaces of a 7 inch radius. Batteries shall conform to the curvature of the above cited curved surface and maintain no greater than 3/16 inch distance between the curved surface and entire battery surface while conformed. It shall meet the requirements of 3.5.3 and 3.6.

33. High temperature temporary cutoff devices (3.7.2.4).

Not less than two devices.

34. Redundant safety protection (3.7.2.7).

Batteries shall have a battery management system, no less than two field-effect transistors, a resettable and permanent fuse protecting all of the terminals, connections, and shielding. All three tiers of protection shall have had to fail prior to allowing any potential uncontrolled power flow to the battery cells that would cause an unsafe condition from any signal or power pins on any connector ports or bottom contacts.

35. Nail penetration, battery (3.7.2.9).

Add:

The user activated display shall flash 000 indicating the battery is not in usable condition but should continue to provide degraded operational capability. Charging shall be disabled while the battery is in fault condition.

36. Electromagnetic interference (3.8).

Add:

The battery shall comply with the applicable emissions and susceptibility requirements of MIL-STD-461 Table V, Aircraft, Army, including Flight Line as specified in 4.7.7.

37. End item interoperability (3.10.2).

The battery shall be capable of supplying the required voltage and current as specified in 3.5.2 and 3.5.5 while connected to the intended integrated U.S. Army system(s), defined at contract award and notionally depicted in FIGURE 12.

Add:

38. **Reliability (3.11).**

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The battery shall have a reliability of not less than 468 hours Mean Time Between Essential Function Failure (MTBEFF). A reliability Essential Function Failure is defined as any incident that results in the inability for the battery to perform its essential functions. When tested as specified in 4.7.2.4.1 batteries shall meet the requirements of 3.4.4.1.5, 3.5.2, 3.5.3, 3.5.6, 3.5.8.2, 3.5.9, 3.5.10, 3.5.10.1, 3.5.11, 3.7, and 3.10.2.

VERIFICATION.

39. First article inspection (4.4).
Groups 1, 2, 3, 4, 5, and 6 apply
d. Group 1 (GR1) – Functional tests: increase sample size from twelve (12) samples to sixteen (16) samples.
h. Group 5 (GR5) – Nail penetration tests (battery level and Weapons Characterization tests – samples size of 18.

Replace TABLE X with the following:

Inspection/Test	Req't	Test Method	Samples or Group (GR)															
			GR 1										GR 2	GR 3	GR 4	GR 5		GR 6
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(5 ea)	(8 ea)	(12 ea)	1- 6	7 - 18	(10 ea)
Visual and mechanical	3.4.1	4.7.1.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓
End item interoperability	3.10.2	4.7.6.2																✓
Battery voltage	3.5.2	4.7.2.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Insulation resistance	3.7.2.1	4.7.4.6	✓	✓	✓	✓	✓	✓	✓	✓								
Altitude	3.6.2	4.7.3.4	✓	✓	✓	✓	✓	✓	✓	✓								
Charge acceptance	3.5.10.1	4.7.2.9										✓						
Concurrent charge/discharge	3.5.11	4.7.2.16										✓						
Capacity discharge (initial)	3.5.3	4.7.2.3	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓
Conformability	3.6.17.3	4.7.3.19.2		✓	✓													
Cycle life	3.5.4	4.7.2.4					✓											
Battery storage life	3.6.8	4.7.3.10						✓										
Overcharge/electrical leakage	3.7.2.2	4.7.4.7							✓									
Low temperature discharge	3.6.1	4.7.2.7							✓									
High temperature discharge	3.6.1	4.7.2.8							✓									
Projectile	3.7.2.8	4.7.4.11							✓									
Nail Penetration, battery	3.7.2.9	4.7.4.12														✓		
High rate discharge	3.5.3	4.7.2.5	✓		✓													

NOTES
Each group MUST be tested in the order listed in the table.

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Inspection/Test	Req't	Test Method	Samples or Group (GR)															
			GR 1										GR 2	GR 3	GR 4	GR 5		GR 6
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(5 ea)	(8 ea)	(12 ea)	1- 6	7 - 18	(10 ea)
Retention of charge	3.5.10.2	4.7.2.10	✓	✓	✓	✓												
Pulse discharge	3.5.3	4.7.2.11	✓		✓													
Inrush current	3.5.5	4.7.2.12								✓								
Temperature shock	3.6.3	4.7.3.5	✓	✓	✓	✓												
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓												
Vibration (during test)	3.6.5	4.7.3.7	✓		✓													
Vibration (loose cargo)	3.6.5	4.7.3.7.1		✓		✓												
Chemical resistance	3.6.13	4.7.3.15									✓							
Connector insertions	3.4.4.1.1.2	4.7.1.3	✓			✓												
Flat terminal strength	3.4.4.1.2	4.7.1.4	✓			✓												
USB power port connector(s)	3.4.4.1.5	4.7.2.17	✓	✓	✓	✓												
Immersion, shallow	3.6.6	4.7.3.8.1	✓	✓	✓	✓												
Immersion, deep	3.6.6	4.7.3.8.2										✓						
Transit drop, normal	3.6.7	4.7.3.9.1	✓	✓	✓	✓												
Transit drop, severe	3.6.7	4.7.3.9.2										✓		✓				
Surface friction	3.4.5.6	4.7.1.8		✓	✓													
Reliability	3.11	4.7.2.4.1																✓
Flexibility test	3.6.17.1	4.7.3.19.1		✓	✓													
Over flexibility protection test	3.6.17.2	4.7.3.19.1.1		✓	✓													
Impact	3.6.9	4.7.3.11									✓							
Salt Fog	3.6.11	4.7.3.13								✓								
ESD	3.6.12	4.7.3.14	✓	✓	✓	✓												
Solar radiation	3.6.10	4.7.3.12								✓								
Rapid decompression	3.6.14	4.7.3.16								✓								
Explosive atmosphere	3.6.15	4.7.3.17									✓							
Combined thermal, humidity and altitude	3.6.16	4.7.3.18									✓							
Immersion, shallow (post drop & flex)	3.6.6	4.7.3.8.1	✓	✓	✓	✓								✓				
Immersion, deep (post drop)	3.6.6	4.7.3.8.2										✓						
Display, smart	3.5.8.2	4.7.2.14.2								✓								

NOTES

Each group MUST be tested in the order listed in the table.

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Inspection/Test	Req't	Test Method	Samples or Group (GR)																
			GR 1										GR 2	GR 3	GR 4	GR 5		GR 6
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(5 ea)	(8 ea)	(12 ea)	1- 6	7 - 18	(10 ea)
SMBus	3.5.9.1	4.7.2.15.1	✓	✓	✓	✓												
Full capacity discharge	3.5.3	4.7.2.3	✓	✓	✓	✓												
Concurrent charge/discharge	3.5.11	4.7.2.16	✓			✓												
Extreme low temp. discharge	3.6.1	4.7.3.2		✓		✓												
Extreme high temp. discharge	3.6.1	4.7.3.3	✓		✓													
Battery case vent	3.4.5.3	4.7.1.6										✓						
Short circuit protection	3.7.2.3	4.7.4.8									✓							
High temp. temporary cut-off	3.7.2.4	4.7.4.9		✓	✓													
High temp. permanent cut-off	3.7.2.5	4.7.4.10		✓	✓													
Electromagnetic interference	3.8	4.7.7											✓					
End item interoperability	3.10.2	4.7.6.2												✓				
Charger interoperability	3.10.1	4.7.6.1												✓				
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5													✓			
Weapons characterization	3.7.2.11	4.7.4.13															✓	

NOTES

Each group MUST be tested in the order listed in the table.

40. Group A inspection (4.5.1).

Battery shall be subjected to capacity discharge (initial) and immersion, shallow (modified).

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41. Group B inspection (4.5.2).

TABLE vi. *Group B inspection*

Inspection / Criteria	Requirement Paragraph	Test Method Paragraph	Group B Sample Percentage			
			GR1 25%	GR2 25%	GR3 25%	GR4 25%
High rate discharge	3.5.3	4.7.2.5	✓	✓	✓	✓
Concurrent charge / discharge	3.5.11	4.7.2.16	✓	✓	✓	✓
SMBus	3.5.9.1	4.7.2.15.1	✓	✓	✓	✓
Flexibility test	3.6.17.1	4.7.3.19.1	✓	✓		
Immersion, shallow	3.6.6	4.7.3.8.1	✓	✓	✓	✓
Transit drop, severe	3.6.7	4.7.3.9.2	✓	✓		
Immersion, shallow	3.6.6.	4.7.3.8.1	✓	✓	✓	✓
ESD	3.6.12	4.7.3.14	✓	✓	✓	✓
Temperature shock	3.6.3	4.7.3.5	✓	✓		
Capacity discharge (initial)	3.5.3	4.7.2.3			✓	✓

42. Group C frequency (4.5.3.1).

TABLE vii. *Group C inspection*

Inspection / Criteria	Requirement Paragraph	Test Method Paragraph	Group C Sample Percentage			
			GR1 25%	GR2 25%	GR3 25%	GR4 25%
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓
Capacity discharge (initial)	3.5.3	4.7.2.3	✓	✓	✓	✓
Impact	3.6.9	4.7.3.11	✓	✓		
Salt fog	3.6.11	4.7.3.13		✓		✓
Flexibility test	3.6.17.1	4.7.3.19.1	✓	✓		
Overcharge/electrical leakage	3.7.2.2	4.7.4.7	✓		✓	
High temperature discharge	3.5.3	4.7.2.8	✓	✓		
Low temperature discharge	3.5.3	4.7.2.7			✓	✓
Temperature shock	3.6.3	4.7.3.5	✓	✓	✓	✓
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓
Vibration	3.6.5	4.7.3.7	✓	✓	✓	✓
Immersion, shallow	3.6.6	4.7.3.8.1	✓	✓	✓	✓
Transit drop, normal	3.6.7	4.7.3.9.1	✓	✓	✓	✓
Immersion, shallow (post drop)	3.6.6	4.7.3.8.1	✓	✓	✓	✓
Capacity discharge (final)	3.5.3	4.7.2.3	✓	✓	✓	✓

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TABLE vii. *Group C inspection* - Continued

Inspection / Criteria	Requirement Paragraph	Test Method Paragraph	Group C Sample Percentage			
			GR1 25%	GR2 25%	GR3 25%	GR4 25%
Extreme low temp. discharge	3.5.3	4.7.3.2	✓			✓
Extreme high temp. discharge	3.5.3	4.7.3.3	✓			✓
Battery case vent	3.4.5.3	4.7.1.6	✓			✓
High temperature permanent cut-off	3.7.2.5	4.7.4.10		✓	✓	
Nail penetration, battery	3.7.2.9	4.7.4.12	✓	✓	✓	✓
Cell balance	3.5.1	4.7.2.1	8 cells			

43. Constant potential, current limited (4.6.1).

A constant potential of 16.8 volts or appropriate voltage requested via the SMBus interface shall be applied with current up to 7.5 amperes, to a charge cut off of 750 milli- amperes.

44. Connector insertion test (4.7.1.3).

The mating connector shall have a 6-7 pin configuration.

Add:

45. **Connector pull force (4.7.1.3.1).**

a. Secure the battery and grip the cable 6 to 12 inches from the connector.
b. Apply a static force of between 16.1 and 16.5 pounds to the cable.
c. Verify that the quick disconnect disengages.
d. Perform a visual and mechanical inspection (see 4.7.1.1) and verify that the cable is not damaged.
e. Perform a visual and mechanical inspection (see 4.7.1.1) and verify that the cable is not damaged.

Batteries shall meet requirements in 3.4.4.1.1.1.

46. Flat terminal strength (4.7.1.4).

Add:

After the completion of the 250 application for each flat terminal, the following shall be performed:

a. Apply a not less than 20 lb. force using a 0.200 ± 0.050 inch diameter rod for not less than 10 minutes to the center of the first flat terminal.
b. Repeat step a for the 2nd flat terminal.
c. Repeat step a for the 3rd flat terminal.
d. Repeat step a for the 4th flat terminal.
e. Batteries shall meet the requirements of 3.4.1.
f. Batteries shall be subjected to immersion, shallow (see 4.7.3.8.1).
g. Batteries shall meet the requirements in 3.4.4.1.2.

47. Capacity discharge (4.7.2.3).

a. Use of 4.6.3 is permitted.

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c. 18 W

48. Cycle life test (4.7.2.4).
a. 18 W until cutoff voltage.
b. Discharge at 2 A to cutoff voltage.
c. 125 cycles

49. Reliability test (4.7.2.4.1).
The following test shall be performed. Batteries shall undergo cycles 1, 2, 3 and 4 described herein and accrue a cumulative minimum of **760 hours** of operation with zero reliability failure allowed. In addition, batteries shall complete Cycles 1, 2, 3, and 4 sequentially and accrue a minimum of **250 hours** each. Record the Battery Cycle Count for the batteries prior to conducting this test.

1. Cycle 1:
d. 18 W
j. 30 W
o. 40 W

2. Cycle 2:
d. 18 W
j. 30 W
o. 40 W

3. Cycle 3:
d. 18 W
j. 30 W
o. 40 W

4. Cycle 4:
d. 2 A

50. Charge acceptance (4.7.2.9).
d. Use of 4.6.3 permitted.
f. Store the batteries at 55°C (131°F) ± 1.1°C.

51. Pulse discharge (4.7.2.11).
d. Discharge at 5 A for 5 seconds, rest for 25 seconds.

52. Display, smart (4.7.2.14.2).
e. Default settings are when the digits are facing either front or rear.

Add:
j. Verify in a dark room that the display when illuminated is compatible (does not

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significantly balloon to obscure 10% vision in a 100 Sq Ft room) the view with either an AN/PVS-7 or AN/PVS-14 night vision gear.

k. Verify that the location of the digital display is centered at the top of the battery or within 1.25 inches of center closer to the left side with the button on the display's right side.

53. SMBus (4.7.2.15.1).

Add:

Connect two cables (Cable C and Cable D) in accordance with System Test Configuration1, Figure 15 in series to a system interface devices (SID). Connect the other end of the SID with cables A and B to the current contractual End User Device (EUD) and verify that the SMBus data is accurately displayed on the EUD.

The following test shall be performed:

a. Charge each in accordance with 4.6; use of 4.6.3 is permitted.
b. Discharge batteries at the 18W rate as follows: one battery at −10 ± 2ºC (+14ºF) and one at 55 ± 2ºC (131ºF). Record the SMBus RelativeStateOfCharge and total amp hours during discharge.
c. Plot the two data curves recorded in step b and verify that the RelativeStateOfCharge tracks the amp hour charge within +0/-5%.

Battery shall meet requirements in 3.5.9.

54. Concurrent charge / discharge (4.7.2.16).

Perform the following charge & discharge tests. Verify that no charge condition or battery failure occurs:

Test A. Connect a 100% State of Charge (SOC) battery to a 2 Amp load at pin 1 of the battery Nett Warrior connector. Apply a 16.8 VDC 1 Amp current limited supply at pin 3 of the battery Nett Warrior connector. Allow charge to continue for 2 hours at room temperature. Verify that for 2 hours that the battery does not take in a charge current less than 0.95 Amps at pin 3 and that the battery SOC is not less than 80% (+0/-5%).

Test B. Connect battery from Test A to a 1 Amp load at pin 1 of the battery Nett Warrior connector. Apply a 16.8 VDC 2 Amp current limited supply at pin 3 of the battery Nett Warrior connector. Allow charge to continue for 2 hours at room temperature. Verify that after 2 hours the battery does not take in a charge current greater than 1.0 Amps at pin 3 and that the battery SOC is greater than 90%. Continue charge/discharge conditions for another 4 hours and verify that the battery SOC is at 100% and verify that SMBus current never exceeds 0 Amps.

Test C. Connect a 90% SOC battery to a 0.9 Amp load at pin 1 of the battery Nett Warrior connector. Apply a 16.8 VDC 1 Amp current limited supply at pin 3 of the battery Nett Warrior connector. Allow charge to continue for 2 hours at room temperature. Verify that after 2 hours the battery SOC is equal to or greater than 90%.

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Test D. Connect a 100% SOC battery to a 2 Amp load at pin 1 of the battery Nett Warrior connector. Apply a 16.8 VDC 1 Amp current limited supply at the Charge + flat terminal contact of the battery. Allow charge to continue for 2 hours at room temperature. Verify that for 2 hours that the battery does not take in a charge current less than 0.95 Amps at the Charge + flat terminal contact and that the battery SOC is not less than 80% (+0/-5%).

55. USB power port connector(s) (4.7.2.17).
    a. USB BC1.2 specification.
    b. 1.5 A at 5 VDC currently with an 18 W load.
    d. See FIGURE 11 for tolerance specification.
    e. Verify that USB connectors along with the Nett Warrior Connector do not interfere with each other's individual insertion or removal while wearing Army tactical gloves.

56. Immersion, shallow (4.7.3.8.1).

The test samples shall be fully charged and configured as follows: one battery shall be connected to an 18 W load (step c.) via a 6-6 connector and one battery shall be connected to an 18 W load (step c.) via a 6-7 connector and two batteries shall be in transport configuration (i.e. connector unmated). The USB shall be unmated/uncovered during the test.

57. Transit drop, severe (4.7.3.9.2).
    b. Perform at ambient temperature.
    c. Perform drops 1 – 10 in TABLE XV of MIL-PRF-32383 on unmated batteries.

58. Impact (4.7.3.11).
    b. Place the battery on an Enhanced Small Arms Protective Insert side plate backed by a sandbag.

59. ESD (4.7.3.14).

Contact and air discharge test shall include each contact (including bottom charge contacts, USB connector(s), and main system connector). The battery shall be tested with no load and under an 18 W load. Verify that a load of 18 W can be sustained for 1 minute following the test. Charge the battery via the SMBus and perform the discharge test once with the primary connector and once with the bottom charge contacts.

60. Explosive atmosphere (4.7.3.17).

Batteries shall be configured with a PM CCS GSS approved U.S. Army system defined at contract award and notionally depicted in FIGURE 12 and FIGURE 13.

The following configuration shall be tested:

    a. Configuration 1. (Refer to FIGURE 12) Configuration 1 has the battery under test attached with two system cables to a System Interface Device (SID) and then the SID is connected to an End User Device (EUD) with associated system cables.
    b. Configuration 2. (Refer to FIGURE 13) Configuration 2 has the battery under test attached with two system cables to a System Interface Device (SID) that in turn is

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attached with a cable to a Faraday shielded box. The box shall be terminated to a 6 pin Nett Warrior bulkhead connector that has it shield electrically bonded to the exterior metal case with not more than 10 milliohms impedance between the SID cable shield and the metal case. Internal to the metal case there shall be a 50 Ohm impedance matched circuit for USB data (+/- 10%) on pins 4 and 5 (USB Data + and USB Data -); 2.5 Ohms (+/- 10%) to ground between pins 3 and 2 (+5 VDC and System Ground) representing a 2 Amp load; and 10 Ohms (+/- 10%) representing a 20 to 25 Watt load (voltage of battery dependent). Note that the Breakout Plug is the same as the mating connector to the CWB. Refer to INFORMATION TABLE A - NETT WARRIOR CONNECTOR PINOUT DESCRIPTION. Coming from the Faraday Caged Breakout Box shall be electrically isolated data for monitoring and recording the voltage/current and noise on each of the three internal loads (50, 2.5, and 10 ohms).

Add:

61. **Flexibility (4.7.3.19.1).**

The following tests shall be performed. The test shall be performed with a Government approved flexure test frame (see FIGURE 5). Charge battery in accordance with 4.6; use of 4.6.3 is permitted.

a. Soak the battery for not less than four hours at not more than -20°C. Completion of steps b to e shall be completed within 30 minutes of removal of the battery from the temperature chamber.
b. Place the battery in the fixture in a manner that allows the center of the battery to bend while holding its edges.
c. Provide sufficient force to flex the battery up and down 1 inch in each direction 200 times.
d. Rotate the centerline of the battery 90 degrees and replace it in the fixture.
e. Repeat step c.
f. Soak the battery for not less than four hours at not less than 60°C.
g. Repeat steps b through e within 30 minutes of removal from the temperature chamber.
h. Perform the display, smart test of 4.7.2.14.2.
i. Perform the visual and mechanical test of 4.7.1.1.

Batteries shall meet the requirements of 3.5.3.

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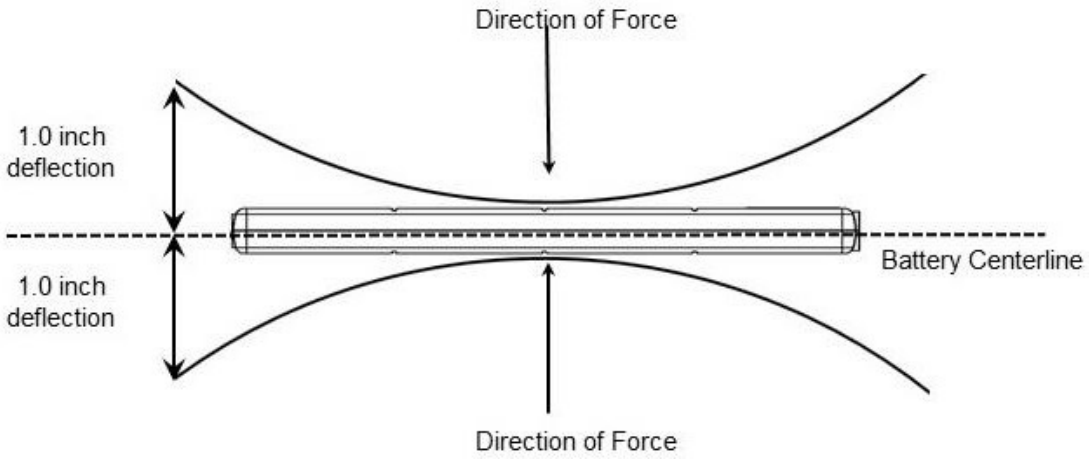


FIGURE 5. Battery flexibility

NOTES
Figure not to scale

Add:

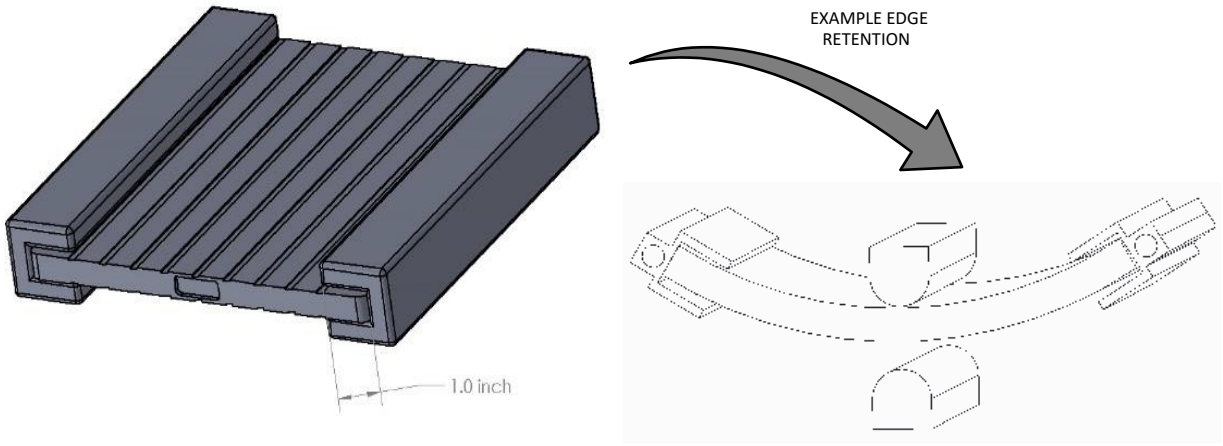


FIGURE 6. Battery edge retaining (reference only)

NOTES
Figure not to scale – pivot and attachment intentionally not shown. The edges can rotate and deflect in the opposite direction of the applied force. The fixture battery edge retention may move inwards when force is applied as necessary to keep the battery in the fixture.

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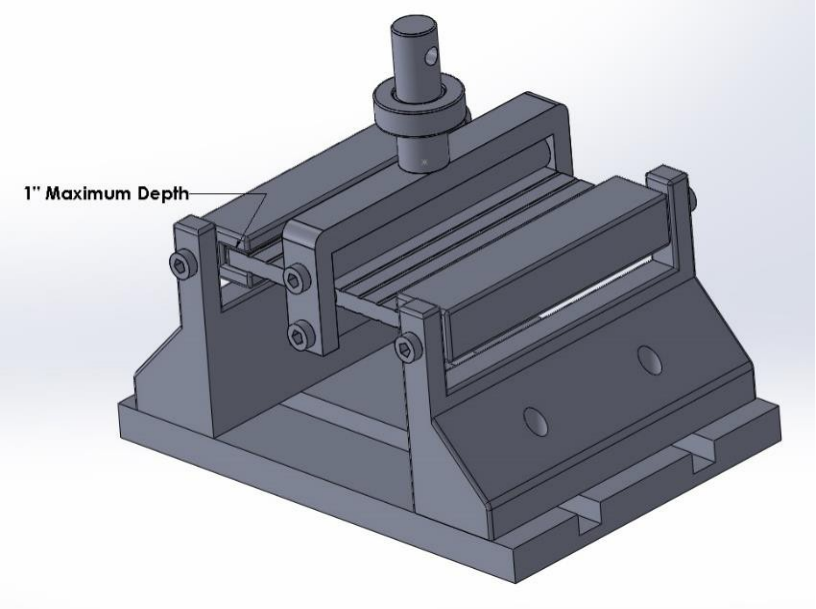


FIGURE 7. Example of over flexibility fixture

NOTES

Figure not to scale. A 1 inch (+/- 0.05”) diameter round bar that extends at a minimum the full axial length of the battery shall be used.

Add:

62. **Over flexibility protection test (4.7.3.19.1.1).**

The following test shall be performed. Charge battery in accordance with 4.6; use of 4.6.3 is permitted. The following tests shall be performed using a Government approved load test frame (see FIGURE 5, FIGURE 6, and FIGURE 7).

a. Soak the battery for not less than four hours at not more than -20°C. The tests steps b to e shall be performed within 30 minutes of removal from the temperature chamber.
b. Place the battery in the fixture in a manner that allows the center of the battery to bend while holding its edges at a maximum of 1.0 inch depth along the entire battery length (see FIGURE 9).
c. Repeat step c with an upward force.
d. Repeat steps c and d for a total of 5 flexes in each direction.
e. Rotate the centerline of the battery 90 degrees and place it in the fixture.
f. Repeat steps c through e above.
g. Soak the battery for not less than four hours at not less than 60°C.
h. Repeat steps b through g within 30 minutes of removal from the temperature chamber.

Battery shall meet the requirements of 3.6.17.2.

Add:

63. **Conformability (4.7.3.19.2).**

An Electrical Mechanical (EM) load frame equipped with a reversible load cell (capacity need not exceed 200 lb-f) shall be used to measure the compressive force required to conform the

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battery to the 7 inch radius curved surface (see FIGURE 8). Ensure that the batteries are un-distressed as they would be provided for shipping.

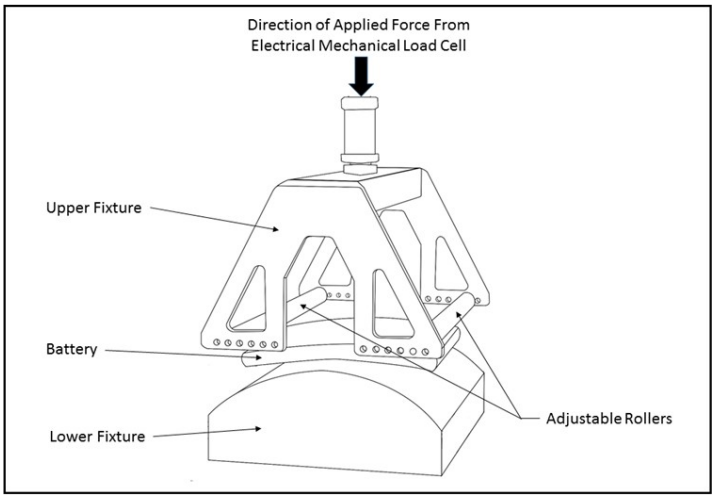


FIGURE 8. Conformability test fixture

NOTES
Figure not to scale.

Note: The rollers shall roll freely on the battery while the upper fixture moves vertically downward. The rollers shall be spread at a fixed distance when the battery is centered over the lower fixture to minimize friction as it curves the battery.

Note: Center the upper fixture above the lower fixture.

For orientation descriptions, see FIGURE 11. The following test shall be performed:

a. Place each battery on a flat surface between each compressive trial to allow it to return to a flat condition.
b. Place the battery on the center of the lower fixture, oriented with the front side upward and the connector on the left.
c. Move the upper fixture to just above or to touch the battery.
d. Ensure that the rollers are within ½ inch of the edges of the battery (+/- ¼ inch). Allow not greater than 1 minute to start the test.
e. Lower the upper fixture with a compressive cross head rate of 2 inches per minute.
f. Record load, displacement, and time measurements at a data acquisition rate of 100 ms.
g. Stop the test when the battery conforms to the 7 inch radius or the force exceeds the 200 lbf, whichever occurs first.
h. Remove the tension of the upper fixture and lay the battery down on a flat surface for not less than 2 minutes.
i. Repeat steps b though h for a total of 5 times.
j. Repeat steps b through i in each of the other three orientations:
    - Front side up with the connector on the bottom

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- Front side down with the connector on the right
- Front side down with the connector on the top

Batteries shall meet the requirements of 3.6.17.3.

64. Nail penetration, cell (4.7.4.4).
SOC not less than 97%

65. Overcharge/electrical leakage (4.7.4.7).
    c. 2.36 A with a charge voltage limit of not less than 50 volts.
    f. 28 W
    k. USB not required.

66. Single penetration test (4.7.4.12.1).
    d. 30Ω +/- 10% for not greater than two minutes prior to start of test.

67. Multiple nail penetration test (4.7.4.12.2).
    d. 30Ω +/- 10% for not greater than two minutes prior to start of test.

68. Weapons characterization (4.7.4.13).
The following test shall be performed for weapon / bullet impact characterization. Refer to notes in section 6.13.
    b. Orient the battery as shown in FIGURE 9. Batteries shall be placed on top of a ballistic plate, inside a currently fielded pouch, and backed by ballistic clay or a sand bag.
    d. 30Ω +/- 10% for not greater than three minutes prior to start of test.
    e. Threats will be specified at time of contract award (see 6.12.3).

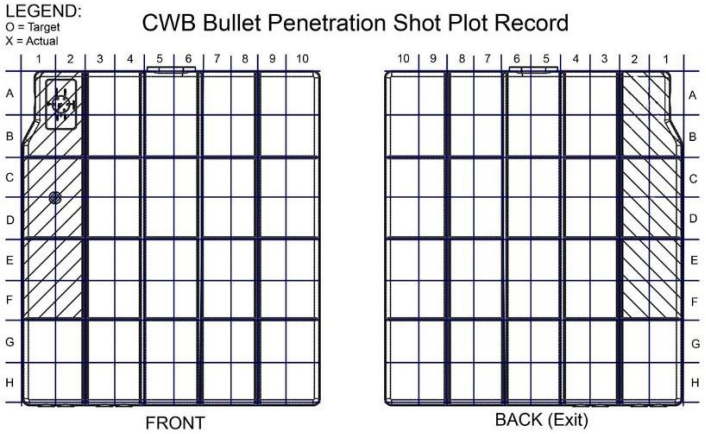


FIGURE 9. Bullet shot location recording template example

Add:

69. **Multiple bullet penetration (4.7.4.14.1).**
Following the procedures in 4.7.4.13, select two areas for bullet penetration in two different

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quadrants of the battery (see FIGURE 10). The batteries under this test shall be recorded using videotape as called out in 4.7.4.13, but with 10 minutes post the 2nd shot. Battery shall meet the requirements of 3.7.2.11.

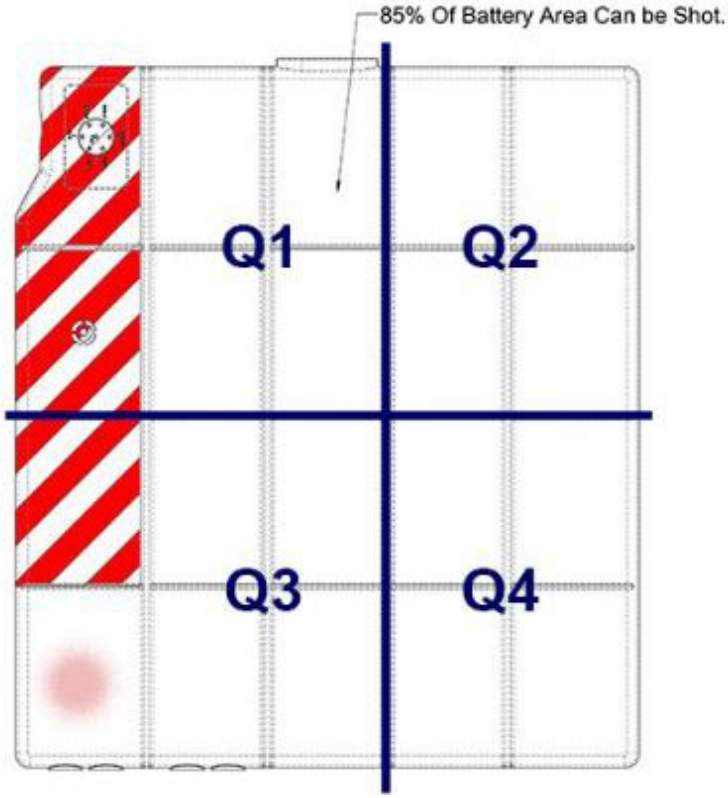


FIGURE 10. Battery quadrants example

70. End item interoperability (4.7.6.2).

Verify that the battery can communicate its status over SMBus to the U.S. Army integrated system(s) defined at contract award and notionally depicted in FIGURE 12.

1. Connect the battery to the input port on the SID.
2. Connect a 7W load to a SID port.
3. Connect an EUD to a port on the SID.
4. Verity that the battery information such as battery SOC, State of Health, and Run-time to empty can be displayed on the EUD by a power monitoring application.
5. Verify the information displayed on the EUD matches that displayed on the battery State-of-Charge/State-of-Health display.

71. Electromagnetic interference (4.7.7).

Battery electromagnetic interference (EMI) shall be characterized with a PM IVAS GSS approved U.S. Army system defined at contract award and depicted notionally in FIGURE 12 and FIGURE 13. Testing shall be conducted in accordance with MIL-STD-461. Tests shall reflect battery contributions to MIL-STD-461 requirements and shall be recorded and verified to not exceed that of MIL-STD- 461 limits.

There are two configurations (see FIGURE 12 and FIGURE 13) for EMI testing. The box shall

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be terminated to a 6 pin Nett Warrior bulkhead connector that has it shield electrically bonded to the exterior metal case with not greater than 10 milliohms impedance between the SID cable shield and the metal case. Internal to the metal case there shall be (see TABLE ii):

1. 50 Ohm impedance matched circuit for USB data (+/- 10%) on pins 4 and 5
2. 2.5 Ohms (+/- 10%) between pins 3 and 2
3. 10 Ohms (+/- 10%) between pins 1 and 2

Monitor and record the voltage/current and noise on each of the three internal loads. Monitoring shall allow for at least 1 GHz sampling at each load. Refer to MIL-STD-461 Bandwidths, TABLE II – Bandwidth and measurement time and Frequency scanning TABLE III – Susceptibility scanning. Configurations and associated MIL-STD-461 Tests are summarized below.

MIL-STD-461 Test	Configuration 1 (see FIGURE 12)	Configuration 2 (see FIGURE 13)
CE102	-	✓
CS101	-	✓
CS116	-	✓
CS118	-	& a Solo Test
RE102	✓	✓
RS101	✓	✓
RS103	✓	✓

CE101; CE105; CS103; CS104; CS105; CS109; CS114; CS115; CS117; RE101; RE103; RS101, and RS105: Are not applicable.

CE102, conducted emissions, radio frequency potential, power leads: Use limits of Figure CE102-1, BASIC CURVE Limit Relaxation.

CS101-1, conducted susceptibility, power leads: Use limits of FIGURE CE101-1 “APPLICABLE CURVE #2” 28 VOLTS OR BELOW.

CS116, conducted susceptibility, damped sinusoidal transients, cables and power leads: Use limits of FIGURE CS116-2. Using the parameters of CS116 conduct test.

CS118-1, personnel borne electrostatic discharge: Use limits of FIGURE CE118-1. Additionally, the Nett Warrior Connector shall be removed from the connected cables and separately tested around its metal perimeter without any other devices connected.

RE102, radiated emissions, magnetic field: Use the limits of FIGURE RE 102-4 (“Navy Mobile & Army” limit with in FIGURE "RE102-4. RE102 limit for ground applications”).

RS103. Testing shall utilize “AIRCRAFT (EXTERNAL OR SAFETY CRITICAL)” level limits for 2 MHz to 18 GHz.

Add:

72. **Basic functionality (4.7.8.1).**

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a. Verify that the capacity of the battery shall be not less than the test requirement in TABLE iii.
b. Verify that the battery takes a charge with a Squad Power Manager (SPM) or Universal Battery Charger-Lite (UBC-Lite) through its Nett Warrior Connector port and that the state of charge increases from 0% to 10% SOC.
c. Verify that the battery takes a charge from a UBC or an Advance Battery Charger through its bottom contacts and fully charges the battery to 100% SOC.
d. Verify that the digital display provides SOC, state of health, run-time-to-empty, charge-time-to-full, and battery cycle count accurately during discharge and charge cycles.
e. Verify that the USB Connector Port(s) are able to provide not less than 1.5 Amps at 5 VDC output.
f. Verify that the battery meets the requirements of 3.4.1.

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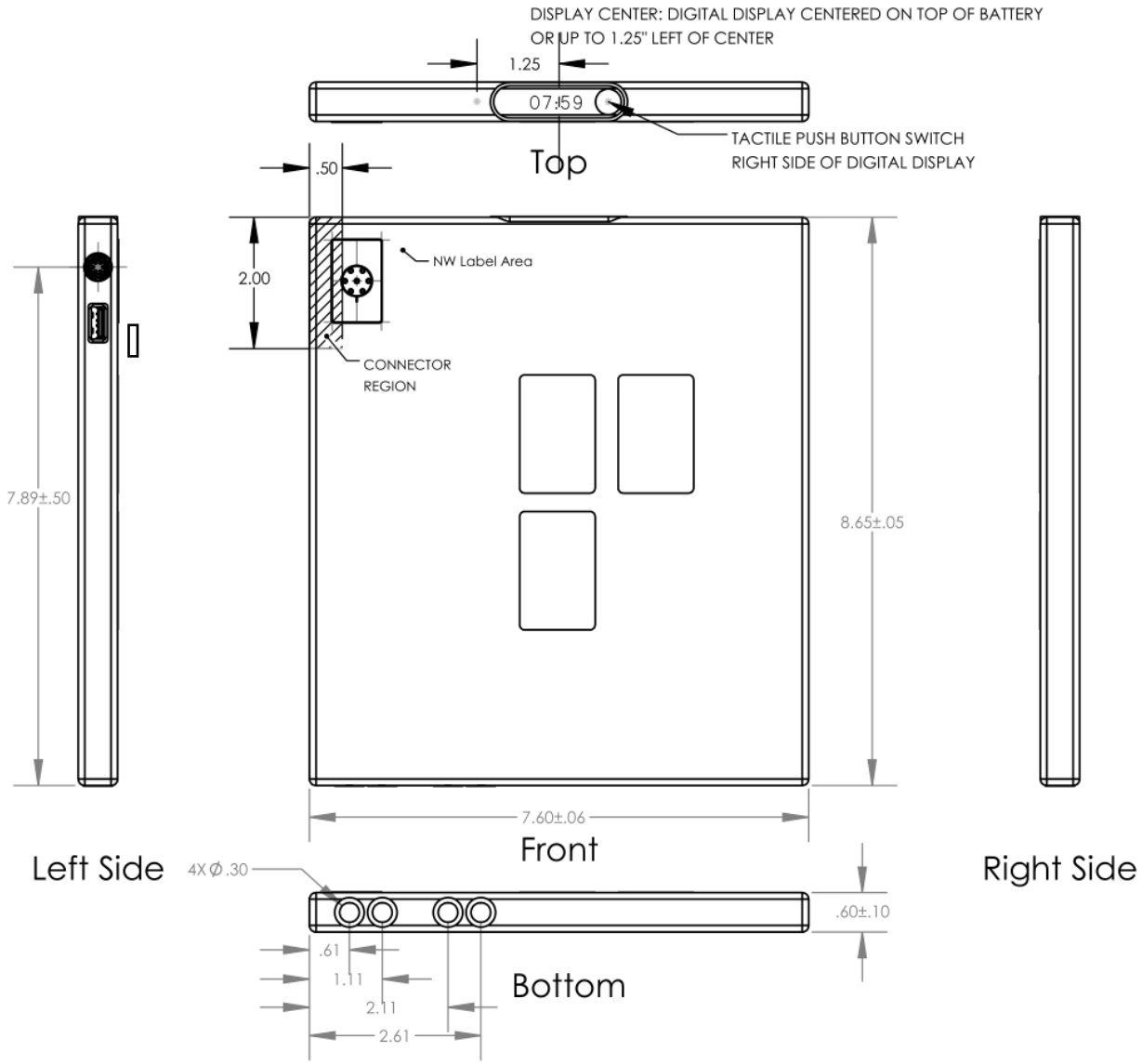


FIGURE 11. Conformal battery dimensions

NOTES

Dimensions are in inches and degrees. Unless otherwise specified, tolerances are:

For distances +/- 0.01 inches

For angles +/- 0.05 degree

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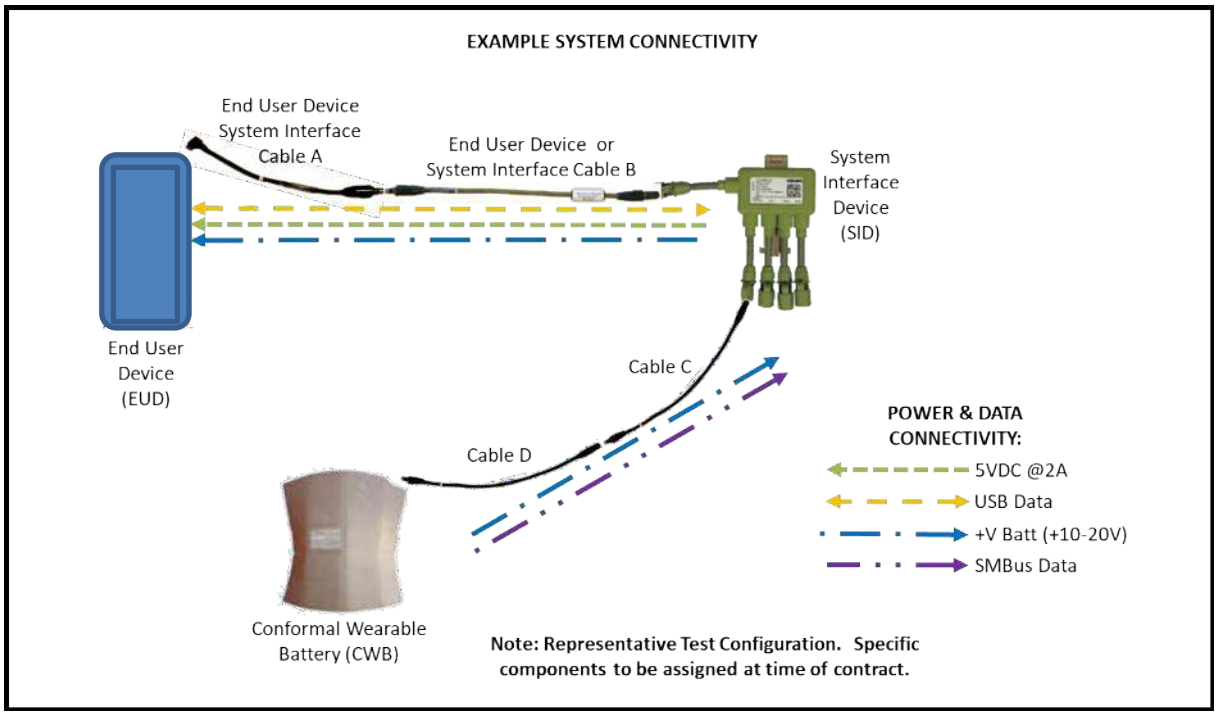


FIGURE 12. System test configuration 1

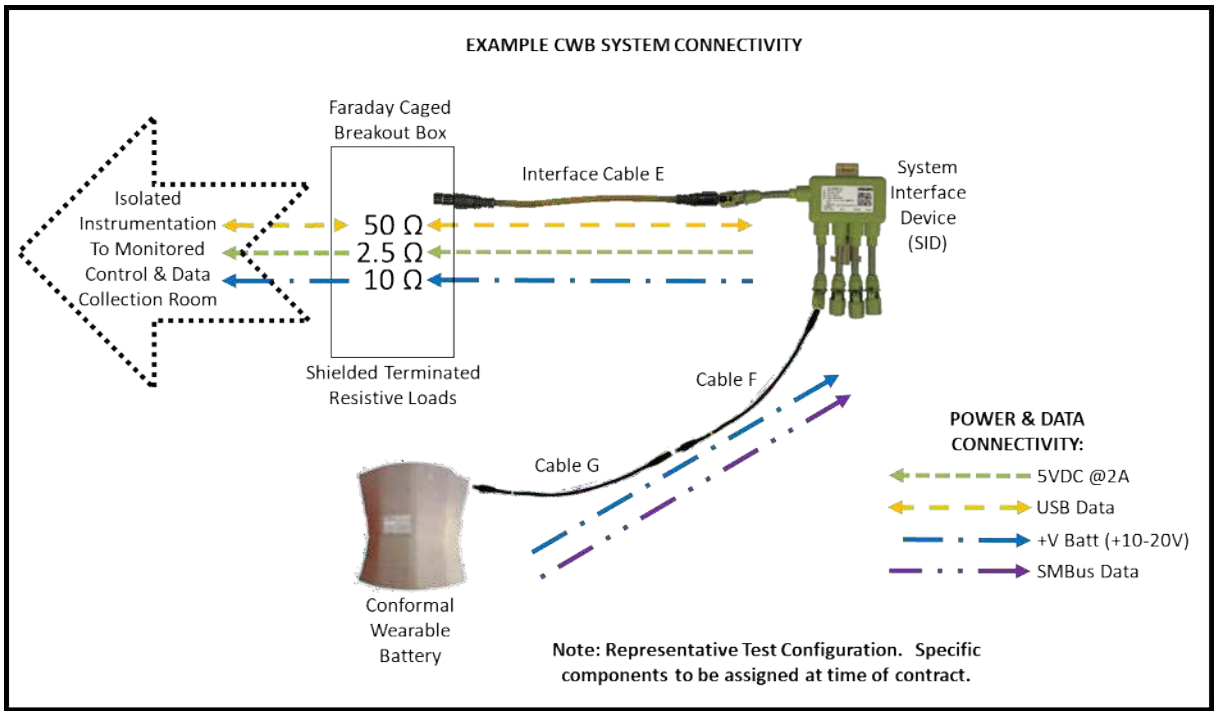


FIGURE 13. System test configuration 2

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NOTES.
(This section contains information of a general nature which may be helpful, but is not mandatory.)

73. Intended use (6.1).
Add:
This battery is intended for use in system such as Nett Warrior, Integrated Visual Augmentation System, and other future integrated U.S. Army systems. The components that comprise the intended system(s) will be specified by the applicable contract. The performance of the battery in and of itself shall not prevent the specified integrated system(s) from complying with the Electromagnetic Interference requirements cited in 3.8 and test methods cited in 4.7.7.

74. Changes from previous issue (6.3).
Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

75. Approved chargers (6.6).
Add:

TABLE viii. *Approved chargers*

ITEM	PART NUMBER	NSN	CAGE
SPM (Protonex/Revision)	SPM-612	6130-01-617-5294	3B0F7
SPM (Protonex/Revision)	SPM-622	6130-01-626-9656	3B0F7
CWB General Adapter (Bren-Tronics)	BTA-70100-910	5940-01-659-4717	51828

Add:
76. **Threats (6.12.3).**
Current threats are known to include:
Muzzle velocity 2390 FPS (+/- 800 FPS), distance between 20 feet to 100 feet from battery.
1. 7.62x54 mm API Tracer Round
2. 7.62x54 mm API Round
3. 7.62x54 mm Trace Round
4. 7.62x54 mm Ball Round
5. 7.62x39 mm API Tracer Round
6. 7.63x39 mm API BZ Round
7. 7.62x39 mm Tracer Round
8. 7.62x39 mm Ball Round (e.g. PS1943)
9. 5.56x39 mm Tracer Round
10. 5.56x39 mm Ball Round

77. Electromagnetic interference (6.16).
Add:
Verification requirements include electromagnetic interference tests of MIL-STD-461 with the battery connected to the system. Battery vendors may need to coordinate with the Project

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Manager, Close Combat Squad (PM CCS), in order to arrange to have their products evaluated in the system. Their address is: Project Manager, Close Combat Squad, ATTN: SFAE-SDR-CCS, 5966 12th Street, Building 1024, Fort Belvoir, VA 22060-5820.

78. Nominal ratings (6.18).
The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes. The following is nominal data for the 148 Watt-Hour Variant:

TABLE ix. *Nominal ratings*

Battery PIN:	**M32383/4-3**
Type Classification:	BB-2525/U
Chemistry:	Li-Ion
Color:	Tan 686A
Weight (max):	2.6 lbs. (1180 grams)
Voltage Range:	10 – 20 V
Nominal Voltage:	14.8 V
Nominal Capacity:	10.0 Ah
Nominal Energy (new battery):	148 Wh at C/5
Battery Life (stored at 30% SOC or greater):	≥ 125 cycles, ≥ 3 years
Rated power output:	≤ 148 W
Continuous load rating:	≤ 5.0 A
Inrush load rating:	35 A (915 µSEC) setting 33 A test; 30 A ((≤25 ms) 25 A Test; 6.4 A (≤500 ms) 6.4 A Test; 5.5 A(≤5000 ms) 5.5 A Test; 5 A Continuous
Charge temperature range:	20 to 50ºC (-4 to 122ºF)
Operating temperature range:	-20 to 50ºC (-4 to 122ºF)
Storage temperature range:	-32 to 71ºC (-27 to 160ºF)
MAX abusive temperature (non-operating):	93ºC (199ºF)
Connector:	Glenair connector Part Number 8070-1939-07ZNU6.7SY or TE Connectivity Part Number: 2828420-2 (7 position)
Overall Dimensions:	8.7 in. x 7.66 in. x 0.70 in.
Non-rechargeable equivalent:	N/A

NOTES
Maximum charge current for connector is 5 A for connector (unless connector supports higher currents) and 5 A minimum with target of >7.5 A minimum for bottom contacts to allow for higher rate charging in the future as chargers improve.

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Custodians:
Army – CR
Navy – MC
Air Force – 71
DLA - CC

Review activities:
Navy – AS, SH

Preparing activity:
Army – CR
(Project Number 6140-2021-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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